The Malaysia Statutory Fiscal-Change Dataset

An LLM-assisted, expert-validated dataset of corporate, personal, and consumption tax changes and discretionary government spending shocks (1990–2023)

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This page accompanies the dataset of **statutory fiscal changes for Malaysia** built for *"Scaling Narrative Fiscal Shock Identification with LLMs"* (Degetau & Samano, World Bank). It is written for the reviewer: it shows the headline picture, lets you download the data, walks several acts in full, and explains how the dataset was produced. It covers both sides of the budget: **statutory tax changes** and **discretionary government spending shocks**.

Each row is **one narratively-announced act × instrument**. On the tax side an instrument is a corporate income tax (CIT), personal income tax (PIT), or broad consumption tax (CONSUMPTION = VAT/GST/SST) change, carrying the statutory rate change. On the spending side it is a discretionary expenditure measure — a stimulus or relief package, a subsidy-policy change, or a large allocation — carrying a magnitude in ringgit rather than a rate. Both carry a motivation label and an exogenous/endogenous read produced with the validated C2b codebook. The unit of observation and the motivation taxonomy follow the narrative approach of Romer and Romer (2010); the LLM pipeline that produces them is validated with the framework of Halterman and Keith (2025). Spending exogeneity additionally carries a preliminary read from the two-condition screen of Das et al. (2026) (non-cyclical motive and no contemporaneous macro rationale), kept alongside the C2b verdict.

! Important

Exogeneity here is a preliminary input, not a finding. The exogenous/endogenous classification is the LLM's read of the narrative record and is provided to *assist* an expert, not to replace one. It is pending expert adjudication. Throughout the project we keep the human in the loop: the model proposes, the expert disposes. The two worked examples below were both chosen because the model's read and the preliminary narrative read disagree — exactly the cases an expert must settle.

1 Statutory tax changes

1.1 Tax changes at a glance

Table 1: **Statutory tax-change inventory.** One row per announced act × tax type. Δpp is the change in the headline statutory rate; motivation and the exogenous/endogenous read are from the validated C2b codebook (v0.9.1).

Act	Effective	Tax	Δpp	Motivation	Exogenous
Corporate income tax rate reduction 34% to 32% (early-1990s Budget)	1994	CIT	−2	Long-run	Exogenous
Corporate income tax rate reduction 32% to 30% (mid-1990s Budget)	1995	CIT	−2	Long-run	Exogenous
Corporate income tax rate reduction 30% to 28% (Budget 1998)	1998	CIT	−2	Counter-cyclical	Endogenous
Corporate income tax rate reduction 28% to 27% (Budget 2007)	2007	CIT	−1	Long-run	Exogenous
Corporate income tax rate reduction 27% to 26% (Budget 2007)	2008	CIT	−1	Long-run	Exogenous
Single-tier corporate tax system (Budget 2008)	2008	CIT	NA	Long-run	Exogenous
Corporate income tax rate reduction 26% to 25% (Budget 2008)	2009	CIT	−1	Long-run	Exogenous
[illegible]	[illegible]	[illegible] CONSUMPTION	[illegible]	[illegible]	[illegible]

Figure 1: **Statutory rate paths by tax type.** Step path of the headline statutory rate implied by the rate-bearing acts. Non-rate acts (one-off levies, regime introductions without a standing rate) are omitted from the path.

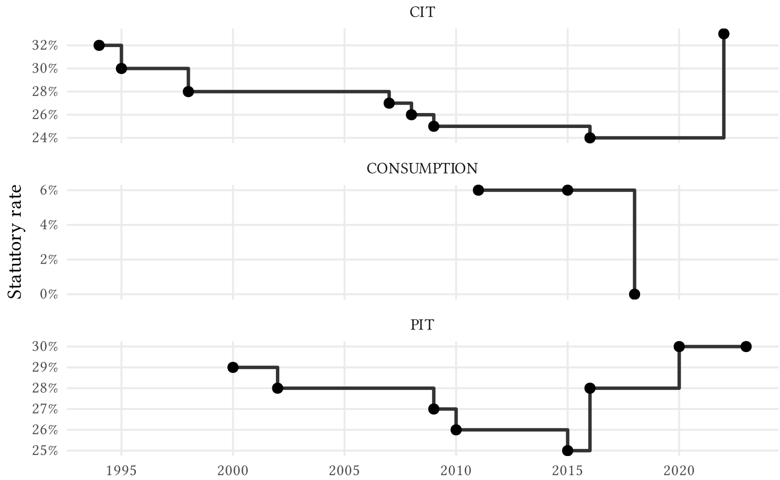
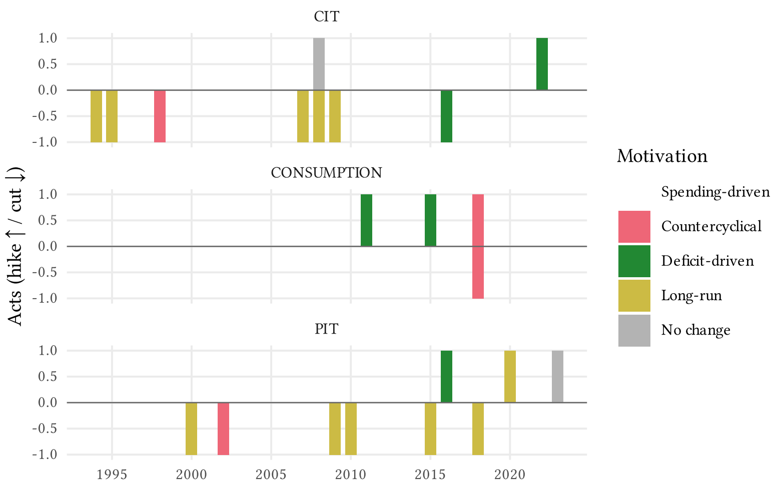


Figure 2: **Tax-change timeline.** Acts per year × tax type as a diverging stacked bar: hikes grow the bar upward, cuts downward. Colour denotes the C2b motivation. Faceted by tax type.



1.2 Download the tax dataset

A clean, flat version of the dataset (one row per act × tax type, no nested provenance columns) is available in three formats. The Stata file carries variable labels; the Excel workbook includes a second sheet with the data dictionary reproduced below.

- Download CSV
- Download Excel (.xlsx, with data dictionary)
- Download Stata (.dta, labelled)

Table 2: **Data dictionary.** Columns of the downloadable dataset.

Variable	Label	Definition
shock_id	Shock identifier	Unique ID, one per announced act x tax type (e.g. MY-CIT-03).
act_label	Act / event name	Human-readable name of the announced act or budget measure.
tax_type	Tax instrument	CIT (corporate income), PIT (personal income), or CONSUMPTION (VAT/GST/SST).
direction	Direction of change	Cut, Hike, or Neutral; consistent with the sign of delta_pp.
rate_from	Statutory rate before (%)	Headline statutory rate prior to the change; NA if not a rate change.
rate_to	Statutory rate after (%)	Headline statutory rate after the change; NA if not a rate change.
delta_pp	Rate change (pp)	rate_to minus rate_from, in percentage points; NA if non-rate.
announced_year	Announcement year	Year the act was announced (budget year).
effective_year	Effective year	Year of assessment / year the change took effect.
effective_quarter	Effective quarter	Effective quarter (e.g. 2016Q1) where known; NA when only year is known.
motivation	Motivation (C2b)	Validated C2b motivation: Spending-driven, Countercyclical, Deficit-driven, or Long-run.
exogenous	Exogenous (C2b)	C2b exogenous/endogenous classification (TRUE = exogenous). Preliminary input pending expert adjudication.

1.3 Two tax acts in full

To show how the dataset is built act-by-act, we reproduce two narrative summaries in the style of Romer and Romer (2010) (their Exhibits 1 and 2, p. 772):

a header with the classification, the identification and motivation reasoning, the most diagnostic source quotation, and links to the source documents. We deliberately picked two acts where the model's classification and the preliminary narrative read **diverge** — the methodologically consequential cases.

1.3.a An endogenous act: the 1998 corporate tax cut

The Budget 1998 corporate tax cut (30% → 28%, announced October 1997) is an **omnibus** budget act: alongside the headline CIT cut it bundled import-duty changes and productivity incentives. Asked for the sign of the effect on tax liabilities, the model returned `"mixed"` — a value outside the permitted `{increase, decrease, no_change}` set, because the package has no single sign. The pipeline degrades gracefully: it keeps the otherwise-valid classification (countercyclical, endogenous) and records the signed effect as `NA`, preserving the raw `"mixed"` for the audit trail. The act also illustrates the preliminary-vs-model tension: read narrowly, the rate cut reads as a long-run competitiveness move (exogenous); read in its Asian-crisis context, the budget reads as cyclical stabilization (endogenous). This is the kind of call we hand to an expert.

Exhibit 1 — Corporate income tax rate reduction 30% to 28% (Budget 1998)

Shock ID: MY-CIT-03 | **Tax:** CIT | **Announced:** 1997 | **Effective:** 1998
Rate change: 30% ⟶ 28% (-2 pp; Cut)
Classification (C2b): Endogenous — Countercyclical
Signed effect on liabilities: NA *(no single sign; raw model output: "mixed")*

Identification. 30->28 reduction announced in the 1998 Budget (Oct 1997), effective YA1998. Evidence: BNM Annual Report 1997.

Motivation (C2b). The evidence consistently shows that the 1998 Budget and accompanying fiscal measures were designed to restore economic stability and return growth to normal in response to the 1997 Asian financial crisis and regional currency turmoil. Policymakers explicitly stated the goal was to 'restore stability to overcome disruptions', 'address imbalances in the economy', and 'restore macroeconomic stability and investor confidence' following the July 1997 currency crisis. While the package included some long-run structural elements (tax incentives for productivity and competitiveness), the predominant and repeatedly stated motivation across all evidence was cyclical stabilization in response to the regional financial shock and its effects on Malaysia's economy (currency depreciation, banking vulnerabilities, current account deficit, inflation). The fiscal measures were explicitly framed as responses to current economic conditions and external shocks, not as structural reforms or deficit reduction from inherited imbalances. This satisfies the countercyclical criterion: actions taken to offset developments that would cause output growth to differ from normal.

> reduce the cost of doing business and strengthen competitiveness … reduced to 28% in 1998 compared with 30% each for Indonesia and Thailand

Preliminary narrative read: exogenous — **differs from the C2b verdict; this is exactly the kind of call an expert must adjudicate.**

Sources: Bank Negara Malaysia Annual Report (1997, en)

1.3.b An exogenous act: the 2000 personal income tax cut

The Budget 2000 across-the-board one-percentage-point cut to personal income tax illustrates the opposite tension. Coming shortly after the 1997–98 crisis, the preliminary narrative read flags it as part of a demand-recovery package (endogenous). The model, weighing the budget's stated rationale — global competition (WTO, AFTA), R&D, human-capital and competitiveness — classifies it as a long-run, exogenous change. Again, the two reads are kept side by side for the expert.

Exhibit 2 — Across-the-board individual income tax cut 1pp (Budget 2000)

Shock ID: MY-PIT-01 | **Tax:** PIT | **Announced:** 1999 | **Effective:** 2000
Rate change: 30% ⟶ 29% (-1 pp; Cut)
Classification (C2b): Exogenous — Long-run
Signed effect on liabilities: decrease (−)

Identification. Headline P1. Budget 2000 across-the-board 1pp cut; framed in BNM AR 1999 as part of the post-1997-98 crisis demand-recovery package -> preliminary endogenous (countercyclical). Top from/to inferred 30->29 (1pp cut applied to the documented 29% pre-2002 top); flagged, rate fields are the inferred top-marginal value.

Motivation (C2b). The 2000 Malaysian Budget's primary stated motivation is to raise long-run growth and improve structural competitiveness, not to offset a cyclical downturn. Policymakers explicitly cite global competition (WTO, AFTA), technological change, low R&D spending relative to developed economies, and the need to improve human resource management and service-sector competitiveness. While consumption-support language appears, the overarching rationale is structural reform and raising potential output through improved incentives and efficiency. The evidence does not characterize current conditions as requiring cyclical stimulus to return growth to normal; rather, it frames the measures as long-run structural improvements to enable Malaysia to compete globally and sustain higher growth.

> one percentage point across-the-board reduction in personal income tax rates, higher tax relief and salary adjustments … real aggregate domestic demand is expected to strengthen (BNM AR 1999, 2000 Budget recovery context)

Preliminary narrative read: endogenous — **differs from the C2b verdict; this is exactly the kind of call an expert must adjudicate.**

Sources: Bank Negara Malaysia Annual Report (1999, en); Budget Speech / Ucapan Bajet (2000, en)

2 Government spending shocks

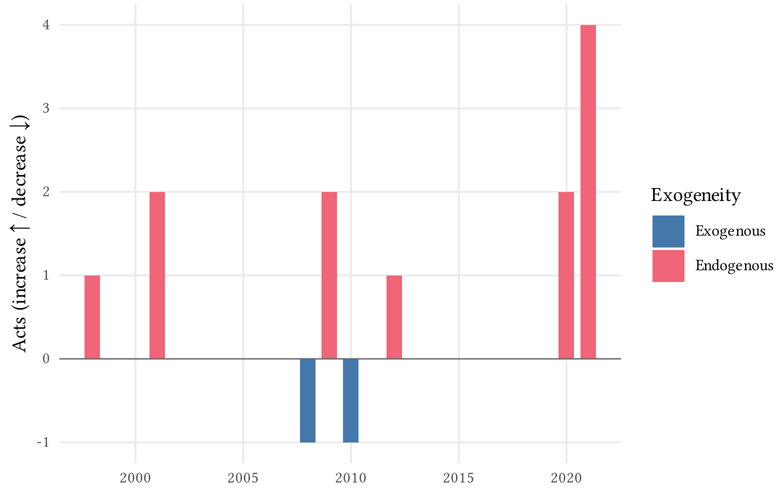
The dataset also covers **discretionary government spending shocks**: major spending actions announced in budgets and crisis responses — stimulus and relief packages, subsidy-policy changes, and large allocations. Unlike a tax change there is no statutory rate, so the magnitude is recorded in ringgit (or as a share of expenditure) as free text rather than as a rate change, and the direction is an increase or decrease in spending. The motivation and the exogenous/endogenous read again come from the validated C2b codebook; a preliminary read from the Das et al. (2026) two-condition screen is carried alongside. Routine baseline development expenditure (for example the standing Five-Year Plan framework) is deliberately excluded — only discretionary changes count as shocks.

2.1 Spending shocks at a glance

Table 3: **Government spending-shock inventory.** One row per announced spending act. The exogenous/endogenous read is from the validated C2b codebook (v0.9.1); category, magnitude, and motivation appear in the exhibit and downloads.

Act	Effective	Exogenous
National Economic Recovery Plan (NERP) (1998)	1998	Endogenous
Pre-emptive fiscal stimulus package, RM3bn (March 2001)	2001	Endogenous
Additional fiscal stimulus package, RM4.3bn (September 2001)	2001	Endogenous
Fuel-subsidy restructuring (June 2008)	2008	Exogenous
First Economic Stimulus Package, RM7bn (November 2008)	2009	Endogenous
Second Economic Stimulus Package, RM60bn (March 2009 mini-budget)	2009	Endogenous
Subsidy rationalisation programme (2010-)	2010	Exogenous
1Malaysia/Keluarga cash-transfer programme (BR1M -> BKM, 2012-)	2012	Endogenous
PRIHATIN Rakyat economic stimulus package (2020)	2020	Endogenous
PENJANA short-term economic recovery plan (2020)	2020	Endogenous
PERMAI assistance package (2021)	2021	Endogenous
PEMERKASA programme (2021)	2021	Endogenous
PEMERKASA+ package (2021)	2021	Endogenous
PEMULIH package (2021)	2021	Endogenous

Figure 3: **Spending-shock timeline.** Acts per year as a diverging stacked bar: spending increases grow the bar upward, decreases downward. Colour denotes the C2b exogenous/endogenous classification. A single panel (no per-category facet): the spending mix is dominated by crisis-response packages.



2.2 Download the spending dataset

The spending dataset is available in the same three formats as the tax dataset (one row per spending act, no nested provenance columns). The Stata file carries variable labels; the Excel workbook includes a second sheet with the data dictionary reproduced below.

- Download CSV
- Download Excel (.xlsx, with data dictionary)
- Download Stata (.dta, labelled)

Table 4: **Spending data dictionary.** Columns of the downloadable spending dataset.

Variable	Label	Definition
shock_id	Shock identifier	Unique ID, one per announced spending act (e.g. MY-SPEND-04).
act_label	Act / package name	Human-readable name of the spending act, package, or policy change.
spending_category	Spending category	Das component family: Infrastructure investment, Social transfers, Subsidies, Public wages, Consolidation/restraint, or Other.
direction	Direction of change	Increase (expansionary), Decrease (contractionary), or Neutral.
magnitude_note	Magnitude (free text)	Headline magnitude in RM or percent terms (e.g. RM60 bn); spending has no statutory rate.
announced_year	Announcement year	Year the act / package was announced.
effective_year	Effective year	Year the spending change took effect.
effective_quarter	Effective quarter	Effective quarter (e.g. 2020Q1) where known; NA when only year is known.
motivation	Motivation (C2b)	Validated C2b motivation: Spending-driven, Countercyclical, Deficit-driven, or Long-run.
exogenous	Exogenous (C2b)	C2b exogenous/endogenous classification (TRUE = exogenous). Preliminary input pending expert adjudication.

2.3 A spending act in full

As on the tax side, we reproduce one spending act in full, again choosing a case where the model's classification and the preliminary narrative read **diverge**.

The June 2008 fuel-subsidy restructuring cut a major subsidy in response to a global crude-oil price spike that had made the subsidy bill fiscally unsustainable. The preliminary Das et al. (2026) screen, seeing a contemporaneous macro trigger, reads it as endogenous; the C2b model, weighing the deficit-reduction rationale, reads it as exogenous and deficit-driven. The two reads are kept side by side for the expert.

Exhibit 3 — Fuel-subsidy restructuring (June 2008)

Shock ID: MY-SPEND-04 | **Category:** Subsidies | **Announced:** 2008 | **Effective:** 2008Q2
Change: Decrease — Retail fuel prices raised ~40.4% in June 2008 as the global crude-oil spike made the subsidy fiscally unsustainable; partial offsets (rice subsidy, fishermen/transport fuel subsidy, motorist fuel cash rebate).
Classification (C2b): Exogenous — Deficit-driven
Signed effect (C2b): decrease (−)

Identification. Distinct from the 2010 structural rationalisation (reviewer decision: separate row). Driven by the 2008 oil-price spike / financing pressure on the subsidy bill -> cyclical/price-driven -> endogenous.

Motivation (C2b). The fuel subsidy restructuring was motivated primarily by fiscal sustainability concerns, not by contemporaneous spending changes or cyclical conditions. Evidence explicitly states the government restructured subsidies to 'avoid prolonged high fiscal deficit' and because subsidy costs had risen unsustainably from RM1.7 billion (2002) to RM7.5 billion (2007), projected to reach RM28 billion in 2008, 'threatening fiscal sustainability.' While mitigation measures (cash rebates, targeted subsidies) were introduced to cushion the impact on citizens, these were secondary responses to the primary deficit-reduction motivation. The action addresses an inherited fiscal burden from past subsidy commitments at volatile global oil prices, not a contemporaneous spending increase or cyclical downturn.

> The fuel subsidy restructuring exercise undertaken by the Government in June caused fuel prices to increase by 40.4%, resulting in a sharp increase in the headline inflation rate, which peaked at 8.5% in July.

Preliminary narrative read (Das screen): endogenous — **differs from the C2b verdict; this is exactly the kind of call an expert must adjudicate.**

Sources: Bank Negara Malaysia Annual Report (2008, en); Economic Report / Tinjauan Ekonomi (2009, en)

3 How the dataset was built

The dataset is the tail of a validated, country-agnostic pipeline. Each stage has its own notebook documenting its construction and validation:

1. **Corpus.** Primary-source fiscal documents (Economic Reports, Budget Speeches, Bank Negara annual reports, 1990–2023) are extracted and quality-checked (corpus verification).

2. **Measure identification (C1).** An LLM codebook surfaces candidate fiscal measures from each document chunk, validated against US ground truth with the Halterman and Keith (2025) framework (C1 evaluation).
3. **Act aggregation (C0).** Surface forms of the same act, scattered across documents and languages, are collapsed into canonical acts (C0 method comparison).
4. **Motivation & exogeneity (C2).** A two-stage codebook extracts motivation evidence (C2a) and classifies each act's motivation and exogeneity (C2b), using the Romer and Romer (2010) taxonomy (spending-driven, countercyclical, deficit-driven, long-run) (cross-country deployment).
5. **Statutory tax-shock identification.** For each tax instrument, an AI-assisted narrative pass reads the surfaced evidence and the source documents directly, recovers misses, and consolidates events at the announced-act grain, with a recall scorecard and a human stamp (CIT · PIT · consumption).
6. **Spending-shock identification.** Because C1 is tax-scoped, the spending side is identified by reading the corpus directly for discretionary spending actions, consolidated the same way with a recall scorecard and a human stamp; preliminary exogeneity uses the Das et al. (2026) two-condition screen (spending identification).
7. **Binding & enrichment.** The frozen per-instrument (and spending) datasets are bound and each act is enriched with the C2b motivation and exogeneity classification (tax deliverable assembly).

The result is the dataset on this page: a transparent, reproducible starting point for expert validation, not a finished verdict. Reviewer disagreements — especially on the exogeneity column — are the intended next input, in keeping with the narrative tradition of Romer and Romer (2010), where every classification is documented so others can check it.

References

Das, Shuvam, Davide Furceri, Nikhil Patel, and Adrian Peralta-Alva. 2026. *AI Meets Fiscal Policy.* {SSRN} {Scholarly} {Paper}. Rochester, NY: Social Science Research Network.

Halterman, Andrew, and Katherine A. Keith. 2025. "Codebook LLMs: Evaluating LLMs as Measurement Tools for Political Science Concepts." *Political Analysis*, September, 1–17.

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